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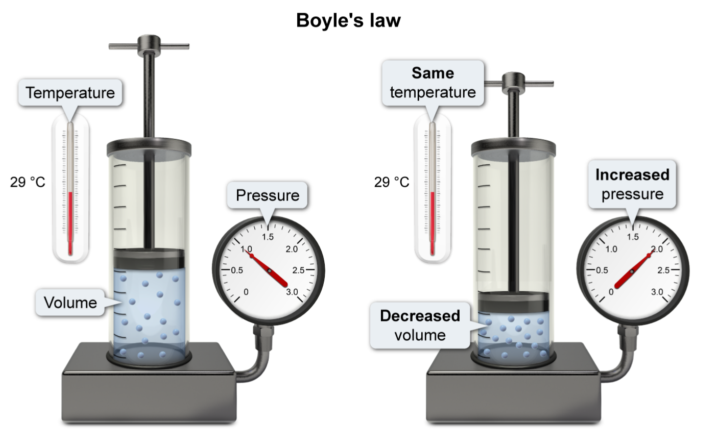
Patients report that their ears "pop" during the pressurization process, similar to being on an airplane in flight. In addition, they report that although there is an initial sense of temperature increasing, the chamber's temperature stays relatively stable during both the pressurization and the depressurization processes.

In a hyperbaric chamber, the volume of intravascular air bubbles decreases as pressure increases. Which of the following gas laws best explains this behavior?

- A. Boyle's law [82%]
- B. Charles' law [13%]
- C. Avogadro's law [1%]
- D. Gay-Lussac's law [3%]

Correct
81% answered correctly

Explanation:



At constant temperature and number of moles of gas, the pressure of an ideal gas is inversely proportional to volume.

According to the kinetic molecular theory of gases, pressure results from the number of collisions that gas particles make with the walls of a container. Decreasing volume while maintaining the same temperature (ie, speed of gas molecules) and the same number of gas particles increases the number of collisions with the walls of the container and increases the pressure. Consequently, pressure and volume are inversely proportional, and every gas has a particular pressure-volume constant at a given temperature.

Boyle's law describes this **inverse relationship** between **volume** V and **pressure** P for a fixed number of molecules of an ideal gas at constant temperature according to the following equations:

$$P \propto \frac{1}{V}$$

$$PV = k$$

where *k* is a constant. These equations show that as pressure increases, volume decreases (and vice versa). Therefore, Boyle's law best explains the decrease in the volume of intervascular air bubbles as the pressure in the hyperbaric chamber increases.

(Choice B) Charles' law describes the directly proportional relationship between the volume and the absolute temperature of a fixed number of molecules of an ideal gas at a constant pressure, where $V \propto T$ and V/T is a constant.

(Choice C) Avogadro's law describes the directly proportional relationship between the number of moles and the volume of an ideal gas at constant temperature and pressure, where $V \propto n$ and V/n is a constant.

(Choice D) Gay-Lussac's law describes the directly proportional relationship between the pressure and the absolute temperature of a fixed number of molecules of an ideal gas kept at a constant volume, where $P \propto T$ and P/T is a constant.

Educational objective:
Boyle's law shows an inverse relationship between the pressure and the volume of a fixed number of molecules of an ideal gas at constant temperature.

Time spent:0 sec
Subject:
General Chemistry

QID:402182
System:
Thermodynamics, Kinetics & Gas Laws

Hyperbaric oxygenation therapy involves placing a patient in a chamber with oxygen-rich air at greater than atmospheric pressure. Normal air has approximately 78% N_2, 21% O_2, and trace amounts of other gases. The pressure within the hyperbaric chamber is typically measured in terms of ATA, or atmospheres absolute (1 ATA is the average atmospheric pressure at sea level, or 1 atm). A hyperbaric chamber set at 2.5 ATA would be the average atmospheric pressure at sea level (1 ATA) plus the additional pressure applied to the chamber (1.5 ATA).

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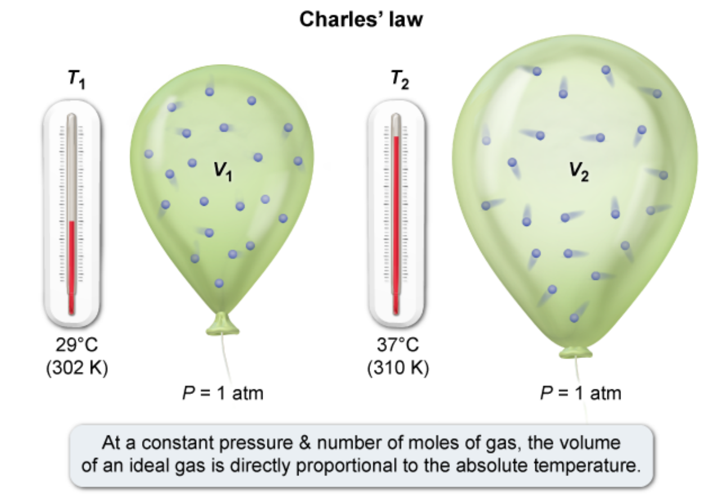
How will the initial volume of inhaled oxygen be affected once it enters the body if the ambient temperature in the chamber is 29 °C and body temperature is 37 °C? (Note: Assume the oxygen is inhaled at a constant pressure.)

- ✔ A. The volume of inhaled oxygen will increase because as temperature increases, volume increases. [88%]
- B. The volume of inhaled oxygen will increase because as pressure increases, volume increases. [2%]
- C. The volume of inhaled oxygen will decrease because as volume decreases, the mass of oxygen decreases. [2%]
- D. The volume of inhaled oxygen will remain the same because the number of moles of oxygen is the same. [6%]

Correct
87% answered correctly

Explanation:



Based on the kinetic molecular theory of gases, the average kinetic energy of the gas particles depends on the temperature. This means that as the temperature increases, the velocity of the gas particles increases. To maintain a constant pressure for a given number of gas particles as the temperature increases, the volume of gas must also increase.

Accordingly, **Charles' law** states that there is a direct relationship between **temperature** *T* and **volume** *V* of a fixed number of particles of an ideal gas at constant pressure, according to the following equations:

$$V \propto T$$

$$\frac{V}{T} = b$$

where *b* is a constant and *T* is the temperature in Kelvin. These equations show that as the temperature increases, the volume of gas also increases.

In this question, the temperature of the oxygen gas that is inhaled increases from 29 °C (302 K) to body temperature, or 37 °C (310 K). This means that the volume of the inhaled oxygen increases when it is inside the body.

(Choice B) Oxygen is inhaled at a constant pressure, so the volume of oxygen will not increase due to changes in pressure.

(Choice C) The mass of oxygen inhaled at 29 °C is the same mass of oxygen present inside the body at 37 °C. Because the amount of mass of oxygen does not change, it is NOT directly responsible for the change in volume.

(Choice D) Although true that the number of moles of oxygen is constant, the volume occupied by the moles of oxygen *will* change upon entering the body because of the increase in temperature.

Educational objective:
Charles' law states that there is a direct relationship between the volume and the temperature of a fixed number of molecules of an ideal gas at constant pressure (ie, as temperature increases, volume increases).

Time spent:0 sec
Subject:
General Chemistry

QID:402183
System:
Thermodynamics, Kinetics & Gas Laws

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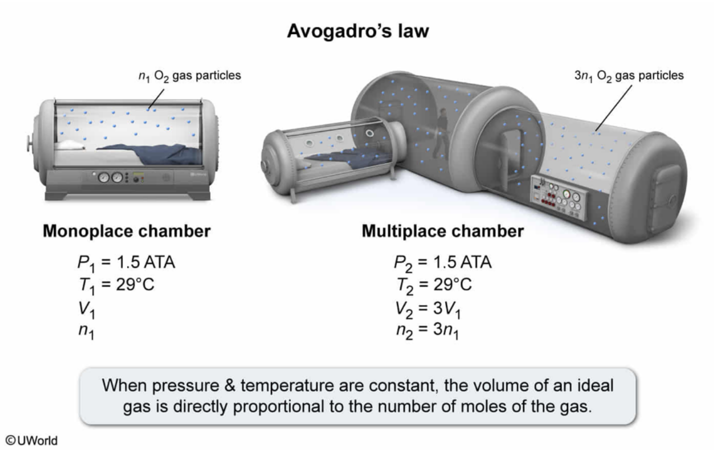
Patients report that their ears "pop" during the pressurization process, similar to being on an airplane in flight. In addition, they report that although there is an initial sense of temperature increasing, the chamber's temperature stays relatively stable during both the pressurization and the depressurization processes.

Suppose that a multiplace chamber has three times the volume of a monoplace chamber, and both are pressurized to 1.5 ATA with 100% O_2 at 29 °C. Compared to that of the monoplace chamber, the number of moles of oxygen in the multiplace chamber would be:

- A. lower. [5%]
- B. the same. [12%]
- C. 2 times higher. [3%]
- ✔ D. 3 times higher. [78%]

Correct
79% answered correctly

Explanation:



The kinetic molecular theory of gases indicates that pressure comes from the combined force of gas particle collisions with the sides of the container. As the number of gas particles increases, the volume must increase for the pressure to remain constant. The identity of the gas(es) within a container does not matter, only the number of moles of gas.

These principles are expressed by Avogadro's law, which states that at a constant temperature and pressure, the **volume** *V* of an ideal gas is directly proportional to the **number of moles** (or particles) *n* of the gas. This is mathematically stated by the following equations:

$$V \propto n$$

$$\frac{V}{n} = d$$

where *d* is a constant. For two different systems under the same temperature and pressure, Avogadro's law can be applied as:

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

where the subscripts indicate the moles and volumes of the two respective systems.

In this question, the multiplace chamber is three times the size of the monoplace chamber, or mathematically, if V_1 is the volume of the monoplace chamber, the volume of the multiplace chamber V_2 is equal to $3V_1$. At a constant pressure of 1.5 ATA, the number of moles of oxygen is directly proportional to the volume:

$$\frac{V_1}{n_1} = \frac{3V_1}{n_2}$$

Rearranging this equation and canceling V_1 gives:

$$n_2 = 3n_1$$

Therefore, the multiplace chamber has three times the number of moles of oxygen compared to the monoplace chamber.

(Choice A) The volume is *directly* proportional to the number of moles of gas. Under the same pressure, larger volumes must have a higher (not lower) number of moles of gas.

(Choice B) The number of moles of oxygen would be the same only if the volumes of the two chambers were the same.

(Choice C) If the ideal gas equation ($PV = nRT$) were used with P = 1.5 atm, V = 3.0 L, R = 0.08 L·atm/mol·K, and T = 29 °C (incorrect units), solving for n would give 1.9 (approximately 2). However, the question *compares* the number of moles of gas in vessels with two *different* volumes under the same conditions.

Educational objective:

Avogadro's law indicates that at constant temperature and pressure, the number of moles of a gas within a container is directly proportional to the volume of the gas.

Time spent: 0 sec	**QID**:402184
Subject:	**System:**
General Chemistry	Thermodynamics, Kinetics & Gas Laws

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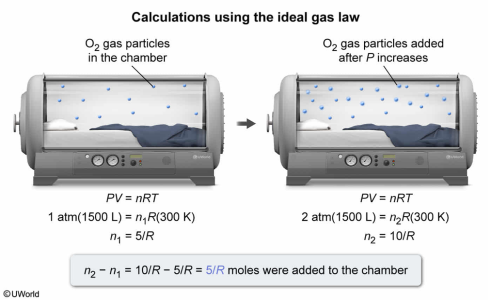
Patients report that their ears "pop" during the pressurization process, similar to being on an airplane in flight. In addition, they report that although there is an initial sense of temperature increasing, the chamber's temperature stays relatively stable during both the pressurization and the depressurization processes.

Assume that the pressure in a 1500 L monoplace chamber is increased from 1 ATA to 2 ATA. Which expression gives the number of moles of oxygen gas *added* to the chamber if the temperature remains constant at 27 °C?

- A. $(4.5 \times 10^5)R$ [4%]
- B. $10/R$ [27%]
- C. $5/R$ [57%]
- D. $0.2R$ [10%]

Correct
57% answered correctly

Explanation:



The ideal gas law combines the relationships described by Boyle's law, Charles' law, and Avogadro's law into a single formula describing the behavior of an ideal gas under a given set of conditions. The ideal gas law relates pressure *P* (atm), volume *V* (L), *n* (moles), and temperature *T* (K) via the **universal gas constant** *R*, which equals 0.08206 L atm mol^{-1} K^{-1}:

$$\mathrm{PV} = \mathrm{nRT}$$

This relationship provides a useful model to approximate the behavior of real gases. However, when doing calculations, it is important to keep the units of measurement consistent with those of *R*.

In this question, the temperature of the monoplace chamber is given in degrees Celsius, but *R* is in terms of Kelvin, so the temperature must be converted to units of Kelvin:

$$27^\circ\mathrm{C} + 273 = 300\ \mathrm{K}$$

In addition, *R* is stated in terms of atm of pressure, but the question gives pressure in units of ATA (atmospheres absolute). According to the passage, 1 ATA equals 1 atm. The *change* in pressure is 2 ATA − 1 ATA = 1 ATA; therefore, this is equivalent to 1 atm.

Substituting the converted values into the ideal gas equation yields:

$$(1\ \mathrm{atm})(1500\ \mathrm{L}) = (n)\left(\mathrm{R}\ \frac{\mathrm{L} \cdot \mathrm{atm}}{\mathrm{mol} \cdot \mathrm{K}}\right)(300\ \mathrm{K})$$

Solving for the number of moles gives:

$$n = \frac{(1\ \cancel{\mathrm{atm}})(1500\ \cancel{\mathrm{L}})}{\left(\mathrm{R}\ \frac{\cancel{\mathrm{L}} \cdot \cancel{\mathrm{atm}}}{\mathrm{mol} \cdot \cancel{\mathrm{K}}}\right)(300\ \cancel{\mathrm{K}})} = \frac{5}{R}\ \mathrm{moles}$$

Therefore, the number of moles of oxygen *added* to the chamber is 5/*R* moles.

(Choice A) This expression is obtained if 1500 is multiplied by 300*R* instead of divided by 300*R*.

(Choice B) This expression is obtained if the final pressure of 2 ATA is used for the calculation instead of the pressure change (added pressure) of 1 ATA (1 atm).

(Choice D) This expression is obtained if 300 is divided by 1500 instead of 1500 divided by 300.

Educational objective:
The ideal gas law describes the number of moles of an ideal gas under given conditions of pressure, volume, and temperature related via the universal gas constant.

Time spent:0 sec
Subject:
General Chemistry

QID:402185
System:
Thermodynamics, Kinetics & Gas Laws

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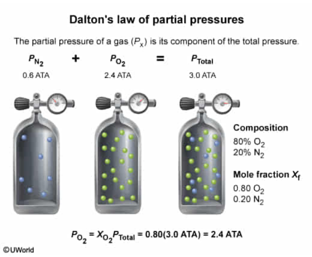
What is the partial pressure of O_2 in a 3-ATA multiplace chamber filled with 80% O_2 and 20% N_2?

- A. 0.2 ATA [1%]
- B. 0.6 ATA [5%]
- ✔ C. 2.4 ATA [91%]
- D. 3.0 ATA [1%]

Correct
90% answered correctly

Explanation:



For ideal gases, the identity of a gas is unimportant for determining the pressure of the gas. In a container with a mixture of gases, the **total pressure** is equal to the **sum** of the partial pressures of each of the gases within the container. This is known as **Dalton's law of partial pressures**.

Because the pressure of an ideal gas is based on the number of moles of gas particles, the partial pressure of a particular gas can be determined using a mole fraction. The **mole fraction** χ is the **ratio of the number of moles** of an individual gas to the **total number of moles**.

$$P_{O_2} = \chi_{O_2} P_{\text{total}}$$

In this question, the mole fraction is given as a percentage. If the chamber contains 80% oxygen, then the mole fraction is 80/100, or 0.80. Substituting the mole fraction and pressure values into the equation for partial pressure gives:

$$P_{O_2} = 0.80(3\text{ ATA}) = 2.4\text{ ATA}$$

Therefore, the partial pressure of O_2 in a 3-ATA multiplace chamber filled with 80% O_2 is 2.4 ATA.

(Choice A) A pressure of 0.2 ATA is obtained if only the mole fraction of N_2 is used rather than multiplying the mole fraction of O_2 by the total pressure.

(Choice B) A pressure of 0.6 ATA is the partial pressure of N_2. This is obtained by using the mole fraction of N_2 (0.20) in the calculation rather than the mole fraction of O_2 (0.80).

(Choice D) A pressure of 3.0 ATA is the total pressure of the chamber rather than the partial pressure of O_2 in a chamber with a mixture of 80% O_2 and 20% N_2.

Educational objective:
Dalton's law of partial pressures states that the total pressure of a mixture of gases is equal to the sum of the partial pressure of each of the individual gases. The partial pressure of an individual gas can be found using the mole fraction of the gas multiplied by the total pressure.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402186
System:
Thermodynamics, Kinetics & Gas Laws

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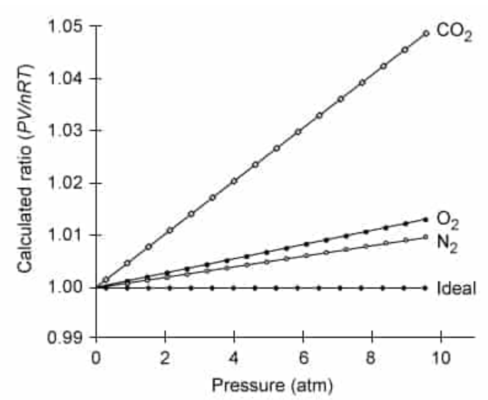
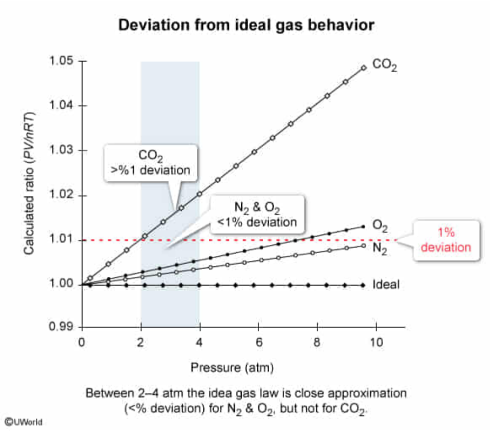
The graph below shows how much certain real gases deviate from ideal behavior. Assume that the ideal gas law gives a close approximation for the behavior of a real gas that has a deviation of 1.0% or less. Based on the graph, which of the following statements is true?



- A. At pressures between 2.0 and 4.0 atm, the ideal gas law gives a close approximation for the behavior of O_2 and N_2, but does not give a close approximation for the behavior of CO_2. [94%]
- B. At atmospheric pressure, the ideal gas law gives a close approximation for the behavior of CO_2 but does not give a close approximation for the behavior of O_2 and N_2. [2%]
- C. At pressures below 4.0 atm, O_2 behaves more like an ideal gas than both N_2 and CO_2.
- D. At pressures above 4.0 atm, the ideal gas law gives a close approximation for the behavior of O_2, N_2, and CO_2. [2%]

Correct
93% answered correctly

Explanation:



Two of the assumptions made by the ideal gas law are that the volume of the individual gas particles is negligible and the individual gas particles do not interact with each other. These assumptions are adequate for most gases at low pressure. However, at higher pressures the individual particles are closer together, resulting in an increased gas **density**.

An increased density results in a greater likelihood that the gas particles will **interact** with each other in an inelastic way. Furthermore, at extremely high pressures, the combined molecular volume occupied by the gas particles becomes **significant** relative to the volume of the container. These factors lead to deviations from ideal behavior in real gases.

To experimentally determine how much a particular gas deviates from ideal behavior, a graph of PV/nRT vs. P can be prepared. Real gases with perfectly ideal behavior should exhibit a constant ratio of PV/nRT equal to 1 because the ideal gas equation states that $PV = nRT$. Carbon dioxide deviates significantly (>1.0%) from ideal behavior at pressures above 2 atm. Oxygen and nitrogen deviate significantly at higher pressures (>8 atm).

Therefore, at pressures between 2 and 4 atm, the ideal gas law gives a close approximation for the behavior of nitrogen and oxygen but does not give a close approximation for the behavior of carbon dioxide.

(Choice B) Atmospheric pressure is defined as 1 atm at standard conditions. Based on the graph, at 1 atm the ideal gas law gives a close approximation for the behavior of *all three* gases.

(Choice C) At pressures below 4.0 atm, the trend lines for the gases show that O_2 has a much lower deviation than CO_2 but a slightly higher deviation than N_2. Therefore, O_2 behaves more like an ideal gas than CO_2, but *not* more so than N_2.

(Choice D) Above 4.0 atm, CO_2 deviates more than 1.0% from ideal behavior; therefore, the ideal gas law does *not* give a close approximation for the behavior of CO_2 above 4.0 atm.

Educational objective:
At higher pressures, deviations from ideal gas behavior occur because the density of a gas increases enough that the individual gas particles are more likely to interact in an inelastic way, and the molecular volume occupied by the gas particles becomes significant relative to the volume of the container.

Time spent:0 sec
Subject:
General Chemistry

QID:402187
System:
Thermodynamics, Kinetics & Gas Laws